

# Assignment

## Problem 1 (15%)

Give the double binary representation of the number +17.3.

## Problem 2 (20%)

Let

$$\beta = 10, \quad p = 4, \quad x = 1.000, \quad y = -0.5555$$

Consider

$$\begin{aligned} x_0 &= x \\ x_1 &= (x_0 \ominus y) \oplus y \quad (*\text{註一}) \\ &\vdots \\ x_n &= (x_{n-1} \ominus y) \oplus y \end{aligned}$$

Assume exactly rounded operations using the rounding up scheme. What are the values of the sequence  $\{x_0, x_1, \dots\}$ ?

## Problem 3 (25%)

Conduct LU factorization with pivoting for the following matrix

$$A = \begin{bmatrix} 4 & -10 & -3 & 7 \\ 6 & -5 & -4 & 7 \\ 6 & 27 & 3 & -3 \\ 12 & 6 & -6 & 6 \end{bmatrix}$$

(a) What are  $M_3, P_3, M_2, P_2, M_1, P_1$  such that

$$M_3 P_3 M_2 P_2 M_1 P_1 = U \quad ?$$

(b) What is

$$P_1^{-1} M_1^{-1} P_2^{-1} P_3^{-1} \quad ?$$

(c) What is

$$P_1^{-1} M_1^{-1} P_2^{-1} M_2^{-1} P_3^{-1} M_3^{-1} \quad ?$$

(d) What is

$$P_3 P_2 P_1 P_1^{-1} M_1^{-1} P_2^{-1} P_3^{-1} \quad ?$$

(e) What is the  $P$  such that

$$PA = LU$$

Is it equal to  $P_3 P_2 P_1$ ?

## Problem 4 (20%)

Consider a binary number in a floating-point system of  $\beta = 2$  and  $p = 6$ :

$$1.11111$$

(1) Transform it to a decimal value with  $\beta = 10$  and  $p = 2$ . That is,

$$x.x$$

Then transform this decimal value to a binary representation with  $p = 7$ . Is the resulting binary the same as (1)?

(2) Redo (1) but a decimal value with  $p = 3$ .

Note that we assume rounding even is used.

## Problem 5 (20%)

In our lecture we derive a version of Cholesky factorization by

$$L_{jj}L_{:j} = A_{:j} - \sum_{k=1}^{j-1} L_{jk}L_{:k}$$

By a similar setting please derive a version that obtains

$$A = U^T U \quad (*\text{註二})$$

You need to give the program.

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### Footnotes:

\*註一:  $\ominus$  為內有減號的圈圈, 即rounded subtraction.

\*註二:  $U^T$  ( $U'$ ) 為transpose of upper triangular matrix  $U$ .